Lesson 1: Zeros, Roots and x-Intercepts

Naming the parts of a parabola, linking the roots of an equation to the zeros of its function and the x-intercepts of its graph, and writing a quadratic function from its zeros.

1. Quadratic Functions and Their Graphs

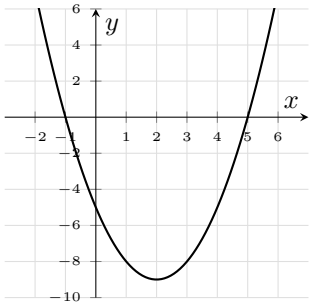
A **quadratic function** can be written as $y = ax^2 + bx + c$ (or $f(x) = ax^2 + bx + c$), where a, b and c are real numbers and $a \neq 0$. This is its **general form**. Its graph is a U-shaped curve called a **parabola**. In function notation, $f(x)$ is the y-value for a given x: $f(3)$ is the value of y when $x = 3$.

- The **axis of symmetry** is the vertical line that splits the parabola into two mirror-image halves. Its equation is $x =$ a number. Because of the symmetry, it is halfway between the two x-intercepts (when it has two).
- The **vertex** is the point where the axis of symmetry meets the parabola: the turning point.
- If $a > 0$ the parabola **opens up**; the vertex is its lowest point and its y-coordinate is the **minimum value** of the function. If $a < 0$ it **opens down**; the vertex is its highest point and its y-coordinate is the **maximum value**.
- The **y-intercept** is where the graph crosses the y-axis ($x = 0$). For $y = ax^2 + bx + c$ it is c. The **x-intercepts** are where the graph meets the x-axis ($y = 0$).

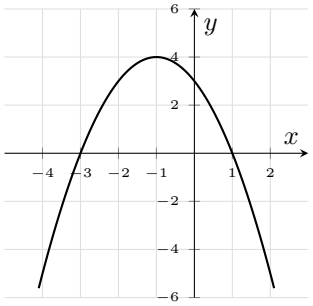
Example 1

The graphs of two quadratic functions are shown. Complete the table.

(a) $y = x^2 - 4x - 5$



(b) $y = -x^2 - 2x + 3$



	(a) $y = x^2 - 4x - 5$	(b) $y = -x^2 - 2x + 3$
Opens up or down		
Vertex		
Axis of symmetry		
Maximum or minimum value		
x-intercepts		
y-intercept		

2. Roots, Zeros and x-Intercepts

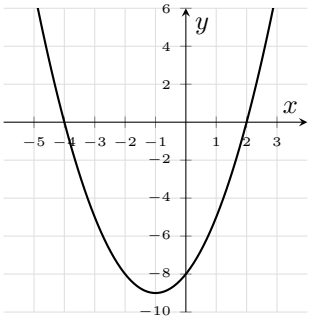
Three words describe the same numbers from three points of view.

- A **root** (or solution) of an *equation* is a value of x that makes the equation true.
- A **zero** of a *function* f is a value of x that makes $f(x) = 0$.
- An **x-intercept** of a *graph* is where the graph meets the x-axis. The point is $(r, 0)$; the number r is also called the x-intercept.

For a function $y = f(x)$: the **zeros** of f = the **x-intercepts** of the graph of $y = f(x)$ = the **roots** of the equation $f(x) = 0$.
Each factor $(x - r)$ of $f(x)$ gives the zero r: $f(x) = (x - 3)(x + 5)$ has zeros 3 and -5.

Example 2

(a) For the linear function $f(x) = 3x - 12$, determine the root of $3x - 12 = 0$, the zero of f and the x-intercept of its graph.



(b) The graph of $f(x) = x^2 + 2x - 8$ is shown. Factor $x^2 + 2x - 8$, then fill in the blanks.

The function f has zeros _____.
The equation $x^2 + 2x - 8 = 0$ has roots _____.
The graph has x-intercepts _____ and y-intercept _____.
(c) How do the factors of $x^2 + 2x - 8$ show the zeros?

3. Finding Zeros and Roots by Factoring

To find the zeros of f algebraically: set $f(x) = 0$, move every term to one side, factor fully, and set each factor that contains x equal to 0.

To find the y-intercept: substitute $x = 0$, i.e. calculate $f(0)$.

The same method works for a product of more than two factors, such as $x(x - 1)(x + 6)$: each factor with an x gives a zero. (Except for a linear equation, never factor before the other side is 0.)

Example 3

Find the roots of each equation.

(a) $x^2 - 5x = 14$

(b) $3(2x + 1)(x - 4) = 0$

(c) $5x^2 - 45 = 0$

(d) $6x^2 + x = 2$

Example 4

For each function, determine the zeros and the y-intercept of its graph.

(a) $f(x) = 2x^2 - 6x - 20$

(b) $g(x) = 4x^2 - 12x + 9$

(c) $h(x) = x^3 - x^2 - 12x$

4. How Many Real Roots?

A parabola can cross the x-axis twice, touch it once at its vertex, or miss it completely.

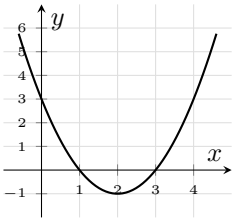
Two x-intercepts $\Leftrightarrow$ two distinct real roots (two zeros). One x-intercept (the vertex is on the x-axis) $\Leftrightarrow$ one real root, a double root. No x-intercepts $\Leftrightarrow$ no real roots (no zeros).

You can often tell from a sketch: a parabola that opens up with its vertex *above* the x-axis (or opens down with its vertex *below* it) never meets the axis.

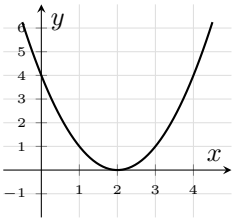
Example 5

(a) The graphs of three functions are shown. For each, state the number of x-intercepts and the real roots of the equation $f(x) = 0$.

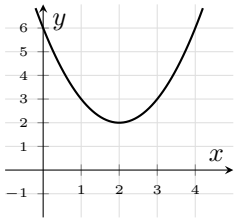
$y = x^2 - 4x + 3$



$y = x^2 - 4x + 4$



$y = x^2 - 4x + 6$



(b) Without graphing, state how many x-intercepts each parabola has.

(i) It opens down; its vertex is $(2, -3)$. (ii) It opens up; its vertex is $(-1, -4)$. (iii) $y = x^2 + 5$

5. Finding Zeros with a Graphing Calculator

Zeros on a TI-84: enter the function in `Y=`, then `GRAPH` (adjust the `WINDOW` until every x-intercept shows). Press `2nd` [`CALC`] `2:zero`; move to the left of one x-intercept and press `ENTER` (left bound), move to its right and press `ENTER` (right bound), then `ENTER` again (guess). Repeat for each x-intercept. (Casio: `G-SOLV`, `ROOT`.)

A decimal such as $1.3333\ldots$ is probably the fraction $\frac{4}{3}$ (on a TI-84, `MATH 1:▶Frac` converts it). A zero r gives a factor $(x - r)$, and a zero $\frac{m}{n}$ gives a factor $(nx - m)$; expand to check that the leading coefficient matches. When the zeros are not rational, the calculator gives decimals: round as the question asks.

Example 6

(a) Use a graphing calculator to find the exact zeros of $f(x) = 6x^2 + 7x - 20$. Use them to write $f(x)$ in factored form.

(b) Find the zeros of $y = x^2 - 3x - 5$ to the nearest hundredth. Can it be factored over the integers?

6. Writing a Quadratic Function from Its Zeros

If a quadratic function has zeros r and s, it has factors $(x-r)$ and $(x-s)$. Every function $y = a(x-r)(x-s)$, $a \neq 0$, has these zeros: changing a stretches or flips the parabola but keeps its x-intercepts. One more point on the graph fixes a.

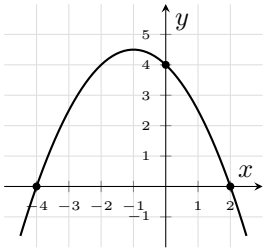
Factored form: $y = a(x-r)(x-s)$ has zeros r and s. To find a, substitute the coordinates of another point on the graph and solve.
A double zero r (the graph touches the x-axis at r) gives $y = a(x-r)^2$.

Example 7

Write each quadratic function in factored form, then in general form.

(a) zeros -2 and 4; the graph passes through $(0, -16)$

(b) zeros 1 and 5; the graph passes through $(3, 6)$



(c) Write the function shown in the graph. The marked points have integer coordinates.

(d) The graph touches the x-axis only at $x = 3$ and passes through $(1, 8)$.

Example 8

A signal flare is fired upward from the top of a cliff. Its height above the sea after t seconds is $h(t) = -5t^2 + 25t + 30$ metres.

(a) Determine the zeros of h. Which zero answers "when does the flare reach the sea?"

(b) Find $h(0)$ and explain what it means.

Lesson 2: Graphing $y = a(x - p)^2 + q$

Seeing how a, p and q stretch, reflect and translate the graph of $y = x^2$, and sketching a parabola from its vertex with the step pattern.

1. The Graph of $y = x^2$

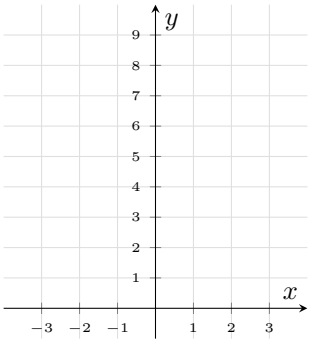
Every quadratic function's graph is the graph of $y = x^2$ moved, stretched or flipped. A quadratic function in **vertex form** is $y = a(x - p)^2 + q$, $a \neq 0$. In this lesson we see what each of a, p and q does to the **base graph** $y = x^2$.

The **domain** of a function is the set of all its x-values, and the **range** is the set of all its y-values. We write them in set notation: $\{x \mid x \in \mathbb{R}\}$ means "all real numbers x", and $\{y \mid y \geq 0, y \in \mathbb{R}\}$ means "all real numbers y that are at least 0".

Example 1

(a) Complete the table of values for $y = x^2$ and graph the function on the grid.

x	-3	-2	-1	0	1	2	3
y							



(b) State each feature of the graph.

Vertex: _____ Axis of symmetry: _____

Minimum value: _____ Domain: _____

Range: _____

(c) Starting at the vertex and moving 1 unit right at a time, by how much does y increase each time?

2. Translations: $y = (x - p)^2 + q$

The parameter q. In $y = x^2 + q$, the number q is added to every y-value, so the graph moves q units **up** ($q > 0$) or **down** ($q < 0$). The vertex moves to $(0, q)$.

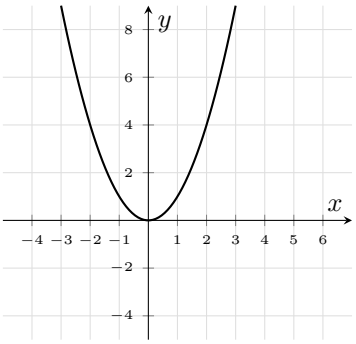
The parameter p. In $y = (x - 3)^2$, $y = 0$ when $x = 3$, $y = 1$ when $x = 2$ or 4, and so on: every y-value of $y = x^2$ now happens 3 units further *right*. So $y = (x - p)^2$ moves the graph p units **right** ($p > 0$) or **left** ($p < 0$). Watch the sign: $y = (x + 3)^2 = (x - (-3))^2$ has $p = -3$ and moves the graph 3 units *left*.

$y = (x - p)^2 + q$ is the graph of $y = x^2$ translated p units horizontally and q units vertically.
Vertex (p, q); axis of symmetry $x = p$; the shape (and so the step pattern $1, 3, 5$) does not change.
If the equation has a constant on the left, as in $y - 2 = (x - 3)^2$, first move it right: $y = (x - 3)^2 + 2$.

A translation does not change the domain, $\{x \mid x \in \mathbb{R}\}$. Since these parabolas open up, the minimum value is q and the range is $\{y \mid y \geq q, y \in \mathbb{R}\}$.

Example 2

The graph of $y = x^2$ is shown.
(a) On the same grid, sketch $y = x^2 - 4$ and $y = (x - 3)^2$ by translating points of $y = x^2$.
(b) State the vertex, the axis of symmetry and the range of each.



Example 3

Describe how each graph is related to the graph of $y = x^2$, and state its vertex and axis of symmetry.

Function	Translation of $y = x^2$	Vertex	Axis
(a) $y = x^2 - 5$			
(b) $y = (x - 4)^2$			
(c) $y = (x + 6)^2$			
(d) $y = (x + 1)^2 + 7$			
(e) $y - 2 = (x - 3)^2$			
(f) $y + 4 = (x + 2)^2 - 1$			

Example 4

(a) Write the equation of the image of $y = x^2$ after a translation of 6 units left.

(b) Write the equation of the image of $y = x^2$ after a translation of 3 units right and 5 units down.

(c) The point $(3, 9)$ is on $y = x^2$. Find its image after a translation of 4 units left and 2 units down, and check that it is on the new graph.

(d) After a translation, the image of $y = x^2$ has its vertex at $(-3, 2)$. Describe the translation and write the equation.

3. Vertical Stretches and Reflections: $y = ax^2$

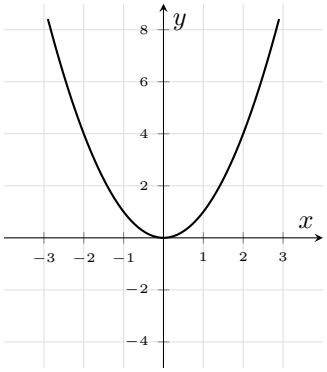
In $y = ax^2$ every y-value of $y = x^2$ is multiplied by a. This is a **vertical stretch by a factor of $|a|$ about the x-axis**: points on the x-axis do not move, and every other point moves away from (or toward) the x-axis.

- $|a| > 1$: the parabola is **narrower** (an expansion). $0 < |a| < 1$: the parabola is **wider** (a compression).
- $a < 0$: the graph is also **reflected in the x-axis**. It opens down and the vertex is a maximum.

Example 5

(a) Complete the table.

x	-2	-1	0	1	2
$y = x^2$	4	1	0	1	4
$y = 2x^2$					
$y = \frac{1}{2}x^2$					
$y = -x^2$					



(b) Sketch the three new graphs on the grid, where $y = x^2$ is shown.
(c) Describe each graph compared with $y = x^2$.

4. Putting It Together: $y = a(x - p)^2 + q$

Compared with $y = x^2$, the graph of $y = a(x - p)^2 + q$ has:
- a **vertical stretch** by a factor of $|a|$ about the x-axis, and a **reflection in the x-axis** if $a < 0$;
- a **horizontal translation** of p units (right if $p > 0$, left if $p < 0$);
- a **vertical translation** of q units (up if $q > 0$, down if $q < 0$).

Vertex (p, q); axis $x = p$. If $a > 0$ it opens up and q is the minimum value; if $a < 0$ it opens down and q is the maximum value.

Order: stretch and reflect first, then translate. Each point (x, y) of $y = x^2$ maps to $(x + p,\ ay + q)$.

When describing a graph, list the stretch, then the reflection, then the translations. (Lesson 3 studies the domain, range and intercepts of these functions.)

Example 6

Describe how the graph of each function is related to the graph of $y = x^2$.

(a) $y = -3(x - 1)^2 + 6$ (b) $y = \frac{1}{2}(x + 5)^2 - 2$ (c) $2y = (x - 8)^2 + 12$

(d) The graph of $y = x^2$ is reflected in the x-axis, stretched vertically by a factor of 4 about the x-axis, and translated 3 units left and 7 units up. Write the equation of the image, and find the image of the point $(2, 4)$.

Example 7

To graph $y = 2(x+4)^2 - 3$ from $y = x^2$, Mila stretches first and then translates. Jonah translates first and then stretches. Follow the vertex $(0, 0)$ and the point $(1, 1)$ through each method. Who is correct?

Mila: stretch by 2, then 4 left and 3 down

Jonah: 4 left and 3 down, then stretch by 2

5. Sketching with the Step Pattern

For $y = x^2$, moving 1, 2, 3 units left or right of the vertex, y goes up 1, 4, 9 (steps of $1, 3, 5$). For $y = a(x-p)^2 + q$ every one of these is multiplied by a.

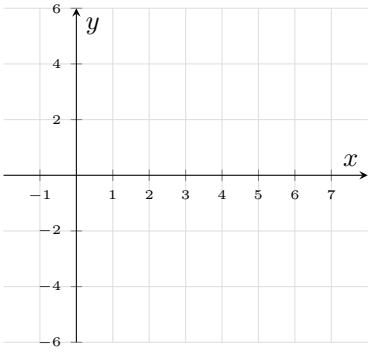
To sketch $y = a(x-p)^2 + q$**:** plot the vertex (p, q). From it, move 1 right (and 1 left) and $a \times 1$ vertically; then 1 more and $a \times 3$; then 1 more and $a \times 5$. Join the points with a smooth U-shaped curve.

For a fraction such as $a = \frac{1}{2}$, move 2 across and $a \times 4$ vertically, so the points land on the grid.

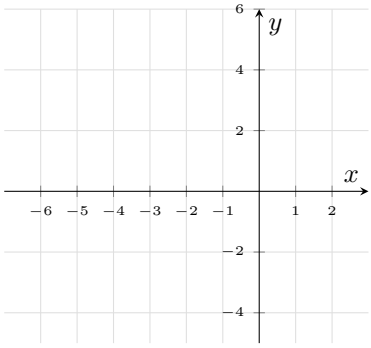
Example 8

Sketch each graph by hand, from the vertex and the step pattern. State the vertex, the axis of symmetry, and the maximum or minimum value.

(a) $y = 2(x-3)^2 - 5$



(b) $y = -\frac{1}{2}(x+2)^2 + 4$



Lesson 3: Characteristics of a Quadratic Function in Vertex Form

Reading the vertex, axis of symmetry, opening, maximum or minimum, domain, range and intercepts straight from $y = a(x-p)^2 + q$.

1. Vertex, Axis of Symmetry and Direction of Opening

In **vertex form** $y = a(x-p)^2 + q$ the key features can be read without graphing. The **vertex** is (p, q) and the **axis of symmetry** is the vertical line $x = p$ through it. The sign of a gives the **direction of opening**, and the vertex is then the lowest or highest point, so q is the **minimum** or **maximum value** of the function.

$y = a(x-p)^2+q$: vertex (p, q); axis of symmetry $x = p$; $a > 0$: opens up, **minimum value** q; $a < 0$: opens down, **maximum value** q.

Watch the sign of p: the bracket is $(x - p)$, so $(x + 3)^2 = (x - (-3))^2$ gives $p = -3$. If the equation is not in this exact form (for example $y + 5 = \dots$ or $2y = \dots$), first solve for y.

Example 1

Complete the table.

Function	Vertex	Axis of symmetry	Opens	Max or min value
(a) $y = 3(x-4)^2 + 1$				
(b) $y = -(x+2)^2 + 6$				
(c) $y = \frac{1}{2}x^2 - 7$				
(d) $y = -2(x-5)^2$				
(e) $y = 0.4(x+3)^2 - 2.5$				

Example 2

Write each equation in the form $y = a(x-p)^2 + q$. Then state the vertex and the maximum or minimum value.

(a) $y - 3 = -2(x+6)^2$ (b) $3y = (x-2)^2 + 12$ (c) $y = 8 - (x-1)^2$

2. Domain and Range

A quadratic function accepts any x-value, so its **domain** is always $\{x \mid x \in \mathbb{R}\}$. Its **range** starts at the y-coordinate of the vertex and goes up (opens up) or down (opens down).

Domain: $\{x \mid x \in \mathbb{R}\}$. Range: $a > 0$: $\{y \mid y \geq q, y \in \mathbb{R}\}$; $a < 0$: $\{y \mid y \leq q, y \in \mathbb{R}\}$.

Example 3

State the domain and range of each function.

(a) $y = 2(x+1)^2 - 9$ (b) $y = -\frac{3}{4}(x-6)^2 + 4$

(c) $y + 2 = 5x^2$ (d) $y = -(x-3)^2$

3. The y-Intercept and Its Mirror Point

The y-intercept is where $x = 0$: substitute 0 for x and evaluate. Because the parabola is symmetric about $x = p$, the point the same distance on the other side of the axis has the same y-value. The mirror of the y-intercept $(0, c)$ is $(2p, c)$. This gives a third point for a sketch.

Example 4

Determine the y-intercept of each graph and the point that is its mirror image in the axis of symmetry.

(a) $y = 2(x - 3)^2 - 5$ (b) $y = -\frac{1}{2}(x + 4)^2 + 10$ (c) $y = 3(x - 1)^2 + 2$

4. How Many x-Intercepts?

Picture the vertex and the direction of opening. A parabola that opens up with its vertex below the x-axis must cross the axis twice; if its vertex is above the axis it never reaches it. If the vertex is on the axis ($q = 0$) the parabola just touches it once.

Number of x-intercepts of $y = a(x - p)^2 + q$: a and q have **opposite signs**: 2; $q = 0$: 1 (the vertex); a and q have the **same sign**: 0.

Example 5

Without graphing, determine the number of x-intercepts. Explain.

Function	Opens	Vertex is above, on or below the x-axis	Number of x-intercepts
(a) $y = 4(x - 2)^2 - 1$			
(b) $y = -3(x + 5)^2 - 6$			
(c) $y = \frac{1}{2}(x - 7)^2 + 3$			
(d) $y = -(x + 1)^2 + 9$			
(e) $y = 6(x - 4)^2$			

5. Finding the x-Intercepts by Solving $a(x - p)^2 + q = 0$

At an x-intercept $y = 0$. In vertex form x appears only once, inside the square, so isolate the square and take square roots. Remember both signs.

To find the x-intercepts: (1) set $y = 0$; (2) isolate $(x - p)^2$; (3) if $(x - p)^2$ equals a negative number there are no x-intercepts; (4) otherwise $x - p = \pm\sqrt{\ }$, so $x = p \pm \sqrt{\ }$. Give exact answers in simplest radical form, then round if asked.

A graphing calculator check: graph the function and use the **zero** feature (once for each x-intercept).

Example 6

Determine the x-intercepts as exact values. Where an answer is not an integer, also round it to the nearest hundredth.

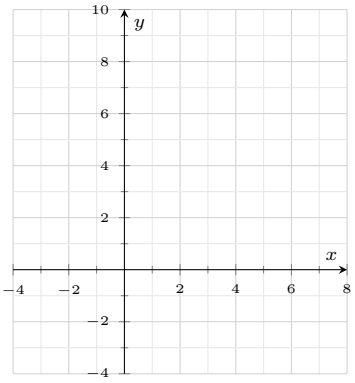
(a) $y = (x-3)^2 - 16$ (b) $y = 2(x+1)^2 - 18$ (c) $y = -3(x-2)^2 + 15$

(d) $y = \frac{1}{2}(x+4)^2 - 6$ (e) $y = 4(x-1)^2 + 3$ (f) $y = -(x+6)^2$

Example 7

Analyze $y = -\frac{1}{2}(x-2)^2 + 8$, then sketch its graph.

Vertex	
Axis of symmetry	
Direction of opening	
Max or min value	
Domain	
Range	
y-intercept	
x-intercepts	



6. Matching Graphs, Symmetry and Contexts

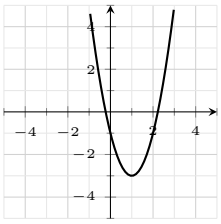
To match an equation to a graph, check the vertex first (careful with the sign of p), then the direction of opening, then the width: a larger $|a|$ gives a narrower parabola. Symmetry also helps: two points with the same y-value are the same distance from the axis, so the two x-intercepts are equally far from $x = p$.

Example 8

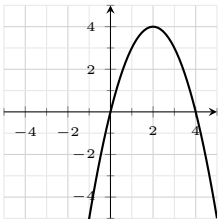
Match each equation with its graph.

A. $y = \frac{1}{2}(x-1)^2 - 3$ B. $y = -(x-2)^2 + 4$ C. $y = 2(x-1)^2 - 3$ D. $y = -(x+2)^2 + 4$

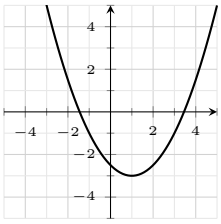
Graph 1



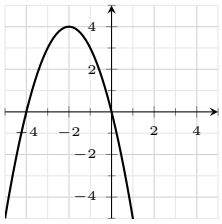
Graph 2



Graph 3



Graph 4



Graph 1: _____ Graph 2: _____ Graph 3: _____ Graph 4: _____

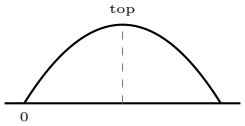
Example 9

(a) The graph of $y = a(x+3)^2 + q$ has an x-intercept at 2. Determine the other x-intercept.

(b) The point $(1, -6)$ is on the graph of $y = 2(x-4)^2 + q$. Name another point on the graph with a y-coordinate of -6.

Example 10

The underside of a footbridge arch is modelled by $h = -0.08(x-10)^2 + 8$, where h is the height in metres above the path and x is the horizontal distance in metres from the left foot of the arch.



(a) What is the maximum height of the arch, and how far from the left foot is it?

(b) How wide is the arch at the path?

(c) How high is the arch 3 m from the left foot?

(d) State the domain and range of h in this context.

Lesson 4: General Form and Factored Form: Intercepts, Axis and Vertex

Finding the zeros, axis of symmetry, vertex, y-intercept, domain and range from $y = ax^2 + bx + c$ and from $y = a(x - r)(x - s)$.

1. Three Forms of a Quadratic Function

The same parabola can be written in three forms. Each form shows some features directly; the others take a little work.

Form	**Equation**	**Read directly**
Vertex form	$y = a(x - p)^2 + q$	vertex (p, q), axis $x = p$, max/min value q
General form	$y = ax^2 + bx + c$	y-intercept c
Factored form	$y = a(x - r)(x - s)$	zeros r and s (the x-intercepts)

In all three forms a is the **same number**: it gives the direction of opening and the shape. To change vertex form or factored form into general form, expand and collect like terms.

Example 1

Write each function in general form and state a, b and c.

(a) $y = 2(x - 3)(x + 1)$

(b) $y = -(x + 4)(x - 2)$

(c) $y = \frac{1}{2}(x - 6)(x + 2)$

(d) $y = 3(x - 1)^2 - 5$

2. Factored Form: Zeros, Axis of Symmetry and Vertex

In $y = a(x - r)(x - s)$, the zero product property gives the **zeros** r and s. The two x-intercepts are mirror images, so the **axis of symmetry** is halfway between them. The vertex lies on the axis: substitute that x-value to get its y-coordinate.

$y = a(x - r)(x - s)$: zeros r and s; axis $x = \dfrac{r + s}{2}$; vertex: substitute $x = \dfrac{r + s}{2}$; y-intercept: substitute $x = 0$.

A factor such as $(2x + 3)$ gives the zero $x = -\frac{3}{2}$ (set the factor equal to 0).

Example 2

For each function, state the zeros, the equation of the axis of symmetry, the vertex, the y-intercept and the maximum or minimum value.

(a) $y = (x-2)(x+6)$

(b) $y = -3(x+1)(x-7)$

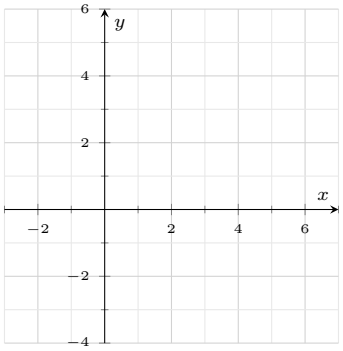
(c) $y = 2x(x-5)$

(d) $y = (2x+3)(x-2)$

Example 3

Analyze $y = -\frac{1}{2}(x+1)(x-5)$, then sketch its graph.

Zeros	
Axis of symmetry	
Vertex	
y-intercept	
Domain	
Range	



3. General Form: the y-Intercept and the Axis $x = -\frac{b}{2a}$

In $y = ax^2 + bx + c$, substituting $x = 0$ gives $y = c$, so the **y-intercept is c**. For the axis of symmetry, compare with $y = ax^2 + bx$: it is the same parabola moved c units up or down, so it has the same axis. Its zeros come from $x(ax+b) = 0$: $x = 0$ and $x = -\frac{b}{a}$. The axis is halfway between them.

$y = ax^2 + bx + c$: y-intercept c; axis of symmetry $x = -\dfrac{b}{2a}$; vertex: substitute $x = -\dfrac{b}{2a}$ into the equation; $a > 0$: minimum, $a < 0$: maximum.

Example 4

Complete the table.

Function	Axis of symmetry	Vertex	Max or min value	y-intercept
(a) $y = x^2 - 8x + 10$				
(b) $y = 2x^2 + 12x + 7$				
(c) $y = -3x^2 + 12x + 1$				
(d) $y = -x^2 - 5x$				

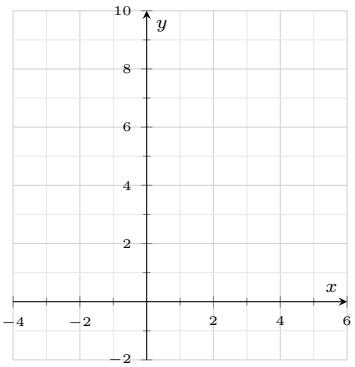
Example 5

Consider $y = -x^2 + 2x + 8$.

(a) State the y-intercept, the axis of symmetry and the vertex.

(b) Write the function in factored form and state its zeros.

(c) State the range, then sketch the graph.



4. When the Zeros Are Not Integers

Many general-form functions do not factor. Two ways to find the x-intercepts:

- **Graphing calculator:** graph the function and use the **zero** feature (once for each x-intercept). This gives decimal answers.
- **Exact values:** find the vertex (p, q) with $x = -\frac{b}{2a}$. Since a is the same in every form, the function is $y = a(x-p)^2+q$. Then solve $a(x - p)^2 + q = 0$ with square roots, as in Lesson 3. (Lesson 6 shows another way to reach vertex form: completing the square.)

Example 6

Consider $y = 2x^2 - 4x - 5$.

(a) Determine the vertex and write the function in vertex form. Check by expanding.

(b) Determine the x-intercepts exactly and to the nearest hundredth. Check with the calculator's zero feature.

5. Using the Best Form

Choose the form that shows what you need: vertex form for the vertex and range, general form for the y-intercept, factored form for the zeros. When an unknown coefficient is involved, use the formula for the axis or substitute a known point.

Example 7

The equations $y = 2(x-1)^2 - 8$, $y = 2x^2 - 4x - 6$ and $y = 2(x+1)(x-3)$ are given.

(a) Show that all three describe the same function.

(b) Complete the table, naming the form that shows each feature most directly.

Feature	Value	Best form
Vertex		
y-intercept		
Zeros		
Range		

Example 8

(a) The graph of $y = x^2 + bx + 11$ has axis of symmetry $x = 3$. Determine b and the vertex.

(b) The function $y = -2x^2 + 8x + c$ has a maximum value of 5. Determine c.

Example 9

A farmer uses 40 m of fencing to make a rectangular pen against a long barn wall (no fence along the wall). If each side touching the wall is x m long, the area in square metres is $A = x(40 - 2x)$.

(a) Write A in the form $a(x-r)(x-s)$ and state the zeros. What do they mean?

(b) Determine the maximum area and the dimensions that give it.

(c) State the domain and range of A in this context.

Lesson 5: Writing a Quadratic Equation from Its Vertex and a Point

Finding a from the vertex and one more point, using the zeros and a point, reading the information from a graph, and modelling arches and flight paths.

1. The Vertex and One Other Point

The vertex (p, q) fixes the position of a parabola, but not its shape: every parabola $y = a(x-p)^2 + q$ has the same vertex whatever the value of a. One more point on the graph fixes a.

To write the equation from the vertex and a point: (1) write $y = a(x-p)^2 + q$ with the vertex values for p and q; (2) substitute the point's coordinates for x and y; (3) solve for a; (4) write the equation with a, p and q (keep x and y as letters).

To get the **general form** $y = ax^2 + bx + c$, expand the square, multiply by a and collect like terms. A **y-intercept** of k is the point $(0, k)$.

Example 1

A parabola has the given vertex and passes through the given point. Write its equation in vertex form, then in general form.

(a) vertex $(3, -5)$, point $(5, 7)$

(b) vertex $(-4, 6)$, point $(-2, -2)$

Example 2

Write the equation of each parabola in vertex form. The value of a is often a fraction: leave it as an exact fraction.

(a) vertex $(6, 1)$, y-intercept 13

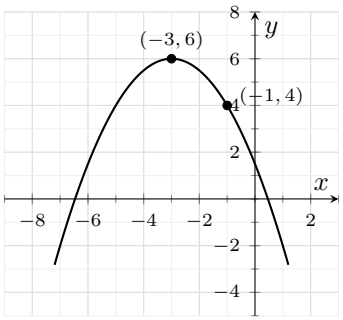
(b) vertex $(-1, 0)$, passing through $(3, -6)$

2. Reading the Vertex and a Point from a Graph

On a graph, read the vertex and one other point where the curve crosses grid lines exactly (a lattice point). Choose a point far from the vertex if you can: a small reading error matters less. Once you have the equation, you can analyze the graph exactly: the y-intercept (set $x = 0$), the x-intercepts (set $y = 0$ and solve with square roots), the axis, the domain and the range. A graphing calculator check: graph your equation and see that it passes through the points you used.

Example 3

The graph of a quadratic function is shown, with its vertex $(-3, 6)$ and the point $(-1, 4)$ marked.



(a) Determine the equation of the function in vertex form.

(b) Determine the y-intercept.

(c) Determine the x-intercepts as exact values and to the nearest hundredth.

(d) State the equation of the axis of symmetry, the domain and the range.

3. Using the Axis of Symmetry

A parabola is symmetric about its axis $x = p$, so points at the same height are the same distance from the axis.

If the axis is $x = p$ and one x-intercept is r, the other x-intercept is $2p - r$ (the axis is midway: $p = \frac{r+s}{2}$). If the equation $y = a(x - p)^2 + q$ has p known but a and q unknown, substitute two points to get two equations, then subtract to eliminate q.

A point whose x-coordinate equals p *is* the vertex.

Example 4

(a) The graph of $y = a(x + 1)^2 + q$ passes through $(1, -2)$ and $(3, 10)$. Determine a and q, and state the minimum or maximum value of the function.

(b) A parabola has axis of symmetry $x = 3$ and one x-intercept at -2. Determine the other x-intercept.

(c) A quadratic function has a minimum value of -12, its axis of symmetry is $x = 4$, and its y-intercept is 20. Determine its equation in vertex form.

4. The Zeros and a Point: Factored Form

When the x-intercepts r and s are known, start from the **factored form** and use one more point to find a.

Factored form: $y = a(x - r)(x - s)$, where r and s are the zeros. Substitute a point (not a zero) to find a. The axis is $x = \frac{r+s}{2}$; substitute it to find the vertex. A graph that only *touches* the x-axis at r has vertex $(r, 0)$: use $y = a(x - r)^2$.

Example 5

Write the equation of each parabola in factored form, then determine its vertex.

(a) zeros -1 and 5, passing through $(3, 12)$

(b) x-intercepts 2 and 6, y-intercept -9

(c) The graph touches the x-axis only at $x = -3$ and passes through $(-1, 8)$.

5. Modelling with a Parabola: Choosing the Origin

Arches, bridge cables, jets of water and the paths of thrown or kicked objects are close to parabolas. To model one:

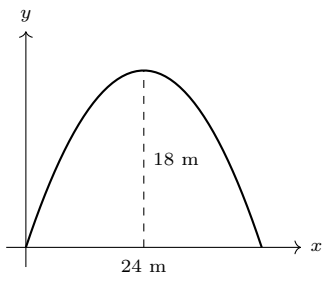
1. Sketch the shape and place the axes. Two good choices: the origin at one end of the base (the zeros are 0 and the width), or the origin on the ground under the vertex (the vertex is $(0, q)$, so $y = ax^2 + q$).
2. Write the vertex and one other point (or the zeros) from the information, and find a.
3. Use the equation to answer the question; state the answer with units and round as asked.

Different origins give different equations, but the same answers to the questions.

Example 6

A stone arch over a garden path is a parabola 24 m wide at its base and 18 m high at its centre.

(a) Place the origin at the left end of the base. Determine an equation for the arch.



(b) How high is the arch 3 m in from one end of its base? Answer to the nearest tenth of a metre.

(c) Write the equation again with the origin on the ground directly under the top of the arch, and use it to check (b).

Example 7

A ball is thrown from a height of 2 m. Its path is a parabola. It reaches its greatest height, 14 m, when it is 10 m away horizontally from where it was released. Place the origin on the ground below the release point.

(a) Determine the equation of the path.

(b) How far away horizontally does the ball land? Answer to the nearest tenth of a metre.

Lesson 6: Completing the Square

Rewriting $y = ax^2 + bx + c$ in vertex form to find the vertex, the axis and the maximum or minimum value.

1. Perfect-Square Trinomials

The general form $y = ax^2 + bx + c$ does not show the vertex; the vertex form $y = a(x - p)^2 + q$ does. **Completing the square** changes general form into vertex form. It is built on perfect-square trinomials:

$$(x + k)^2 = x^2 + 2kx + k^2 \qquad (x - k)^2 = x^2 - 2kx + k^2$$

The constant k^2 is the square of *half* the coefficient of x.

To make $x^2 + bx$ a perfect square, add $\left(\frac{b}{2}\right)^2$: $\quad x^2 + bx + \left(\frac{b}{2}\right)^2 = \left(x + \frac{b}{2}\right)^2$.

Example 1

Determine the number that must be added to make a perfect square, then write the trinomial as a square.

Expression	Add	Perfect square
(a) $x^2 + 12x$		
(b) $x^2 - 18x$		
(c) $x^2 + 7x$		
(d) $x^2 - \frac{2}{3}x$		

2. Completing the Square When $a = 1$

An equation must stay equal, so we cannot just add a number. Instead we add it *and* subtract it (adding 0).

$y = x^2 + bx + c$ **to vertex form:** (1) find $\left(\frac{b}{2}\right)^2$; (2) add and subtract it right after the x-term; (3) write the first three terms as $\left(x + \frac{b}{2}\right)^2$ and combine the constants.
Example: $y = x^2 + 8x + 5 = (x^2 + 8x + 16) - 16 + 5 = (x + 4)^2 - 11$.

Check by expanding the answer: you should get the original general form. A graphing calculator check: graph both forms; the graphs are identical.

Example 2

Write each function in vertex form by completing the square.

(a) $y = x^2 - 6x + 14$ (b) $y = x^2 + 2x - 8$ (c) $y = x^2 + 5x + 3$

Example 3

Complete the square. State the vertex, the axis of symmetry, the minimum value and the range.

(a) $y = x^2 - 10x + 31$ (b) $f(x) = x^2 + 3x + 5$

3. Completing the Square When $a \neq 1$

The method needs x^2 with coefficient 1, so first factor a out of the x^2- and x-terms only (not the constant).

$y = ax^2 + bx + c$ **to vertex form:** (1) factor a from the first two terms: $y = a\left(x^2 + \frac{b}{a}x\right) + c$; (2) inside the brackets, add and subtract the square of half the new x-coefficient; (3) move the subtracted number out of the brackets by **multiplying it by** a; (4) factor the perfect square and combine the constants.

Example 4

Write each function in vertex form.

(a) $y = 2x^2 + 12x + 7$ (b) $y = 4x^2 - 24x + 31$ (c) $y = 3x^2 + 9x - 2$

4. Negative and Fractional Values of a

The same steps work when a is negative or a fraction. Take care with signs: factoring -3 out of $-3x^2 - 12x$ gives $-3(x^2 + 4x)$, and moving -4 out of brackets multiplied by -3 gives $+12$. If the terms are not in order, rearrange them first ($5 - 12x - 3x^2 = -3x^2 - 12x + 5$). For a fraction, dividing by a is the same as multiplying by its reciprocal: $\frac{1}{2}x^2 - 3x = \frac{1}{2}(x^2 - 6x)$.

Example 5

Write each function in vertex form. State the vertex, the direction of opening and the maximum or minimum value.

(a) $y = -x^2 + 8x - 3$

(b) $y = 5 - 12x - 3x^2$

(c) $y = \frac{1}{2}x^2 - 3x + 1$

(d) $y = -\frac{1}{3}x^2 + 2x + 4$

5. Using the Vertex Form

Once a function is in vertex form, you can read the vertex and the maximum or minimum value, and you can solve for the zeros with square roots (set $y = 0$, isolate the square, take $\pm$ the square root). If the square would have to equal a negative number, there are no x-intercepts.

Example 6

(a) Write $f(x) = 2x^2 - 8x - 3$ in vertex form, state its minimum value, and determine its zeros as exact values and to the nearest hundredth.

(b) The height of a ball, in metres, t seconds after it is thrown is $h = -5t^2 + 20t + 1$. Complete the square to find the greatest height and when it occurs.

6. The General Case and Common Errors

Completing the square works with letters too. Common errors to watch for:

- factoring a out of the x^2-term but not out of the x-term (or out of the constant as well);
- forgetting to multiply the subtracted number by a when it leaves the brackets;
- a sign error when a is negative; writing $(x + 3)^2$ for $x^2 - 6x + 9$ (the sign inside matches the sign of b).

Example 7

(a) Complete the square for $y = x^2 + bx + c$, and state the vertex in terms of b and c.

(b) A student wrote $y = -2x^2 + 12x - 5$ in vertex form. Find the step with the error, and correct the work.

Step 1: $y = -2(x^2 - 6x) - 5$
Step 2: $y = -2(x^2 - 6x + 9 - 9) - 5$
Step 3: $y = -2(x - 3)^2 - 9 - 5$
Step 4: $y = -2(x - 3)^2 - 14$

Lesson 7: The Quadratic Formula

Deriving the formula by completing the square, using it to solve any quadratic equation exactly or to a given accuracy, and choosing an efficient method.

1. Deriving the Quadratic Formula

Many quadratic equations, such as $3x^2 - 12x + 7 = 0$, cannot be factored over the integers. Completing the square (Lesson 6) always works: rewrite the equation as $(x - p)^2 = k$ and take the square root of both sides. Doing this once with the letters a, b and c gives a formula that solves *every* quadratic equation.

Example 1

(a) Solve $3x^2 - 12x + 7 = 0$ by completing the square. Give the exact roots in simplest form.

(b) Follow the same steps for $ax^2 + bx + c = 0$, $a \neq 0$. Fill in each line.

Start with the general equation. $ax^2 + bx + c = 0$

Subtract c, then divide every term by a.

Add $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2}$ to both sides.

Factor the left side as a perfect square; write the right side over the common denominator $4a^2$.

Take the square root of both sides ($\sqrt{4a^2} = 2a$; the $\pm$ covers the sign of a).

Subtract $\frac{b}{2a}$ from both sides.

The quadratic formula: the roots of $ax^2 + bx + c = 0$, $a \neq 0$, are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

The whole numerator, $-b \pm \sqrt{b^2 - 4ac}$, is divided by $2a$. The expression $b^2 - 4ac$ under the radical is the **discriminant** (Lesson 8).

2. Solving with the Quadratic Formula

1. Write the equation in standard form $ax^2 + bx + c = 0$ and read a, b and c *with their signs.*
2. Substitute, using brackets for negative values: $(-3)^2 = 9$, but $-3^2 = -9$.
3. Simplify $b^2 - 4ac$ first. If it is negative, there are **no real roots** (a negative number has no real square root).
4. Write the radical in simplest form (Unit 2), then divide out a factor common to *all* the terms: $-b$, the coefficient of the radical, and $2a$.
5. For decimal answers, evaluate the + root and the − root separately with a calculator and round only at the end.

Example 2

Solve. Give exact roots in simplest form and roots to the nearest tenth.

(a) $x^2 + 4x - 3 = 0$ (b) $9x^2 - 6x - 4 = 0$ (c) $x(x + 6) = 2(2x + 7)$

Example 3

Solve $6x^2 + x - 15 = 0$ (a) with the quadratic formula, (b) by decomposition.

The formula always works. Here $b^2 - 4ac = 361 = 19^2$ is a perfect square, so the roots are rational and the equation could also be factored.

Example 4

(a) Determine the zeros of $y = 2x^2 + 6x - 3$, exactly and to the nearest hundredth, and state the x-intercepts of its graph. Check with a graphing calculator: graph the function, then use 2nd [TRACE] (CALC), 2: zero.

(b) Show that $y = x^2 - 2x + 6$ has no zeros. What does this mean for its graph?

3. Finding and Correcting Errors

The most common mistakes with the formula: using b instead of $-b$; squaring a negative b without brackets; a sign error in $-4ac$ when a or c is negative; dividing only one term of the numerator by $2a$; not putting the equation in standard form first.

Example 5

A student solved $2x^2 - 4x - 3 = 0$ as shown. Find the two errors, explain each one, and give the correct roots.

Line 1: $x = \dfrac{-(-4) \pm \sqrt{-4^2 - 4(2)(-3)}}{2(2)}$

Line 2: $x = \dfrac{4 \pm \sqrt{-16 + 24}}{4} = \dfrac{4 \pm \sqrt{8}}{4}$

Line 3: $x = \dfrac{4 \pm 2\sqrt{2}}{4}$

Line 4: $x = 1 \pm 2\sqrt{2}$

4. Choosing a Method

Every method gives the same roots; choose the one that is quickest for the equation in front of you.

Method	**Best when**	**Example**
Factoring	the trinomial factors easily over the integers, or there is no constant term	$x^2 - 3x - 10 = 0$, $4x^2 = 9x$
Square roots	there is no x term, or the equation is $a(x - p)^2 + q = 0$	$5x^2 = 20$, $(x - 1)^2 = 7$
Quadratic formula	it does not factor (or factoring is hard to see); it always works	$3x^2 + 5x - 1 = 0$
Completing the square	you also want the vertex form; it always works, but takes longer	$x^2 + 6x + 2 = 0$
Graphing (zero)	a decimal answer is enough, or to check an algebraic answer	any equation

Example 6

Choose an efficient method, name it, and solve. Give exact roots.

(a) $5x^2 - 45 = 0$ (b) $(x-3)^2 = 12$ (c) $x^2 - x - 20 = 0$ (d) $3x^2 + 5x - 1 = 0$

5. Equations That Need Rearranging First

The formula needs standard form. Expand and collect terms; clear fractions by multiplying every term by the LCD; divide out a common factor to make the numbers smaller. If the variable is in a denominator, state the restriction (e.g. $x \neq 0$), multiply both sides by it, and check that no root breaks the restriction.

Example 7

Write each equation in standard form, then solve. Give exact roots and roots to the nearest tenth.

(a) $(2x+1)(x-3) = (x+2)(x-1)$

(b) $\dfrac{x^2}{3} - \dfrac{x}{2} - 1 = 0$

(c) $3x - \dfrac{2}{x} = 4, \;\; x \neq 0$

(d) $12x^2 - 36x = 15$

6. Problems Solved with the Formula

Set up the equation as in Unit 4, Lesson 5. When it does not factor, use the formula, then reject inadmissible roots (negative lengths or times) and round as the question asks.

Example 8

(a) A ball is thrown upward. Its height after t seconds is $h = -4.9t^2 + 15t + 2$ metres. When does it hit the ground, to the nearest tenth of a second?

(b) A rectangle is 6 cm longer than it is wide, and its area is 24 cm^2. Determine its exact dimensions and exact perimeter, then give each to the nearest hundredth.

Lesson 8: The Discriminant

Using $b^2 - 4ac$ to predict the number and nature of the roots of a quadratic equation and the x-intercepts of its graph, and to find unknown coefficients.

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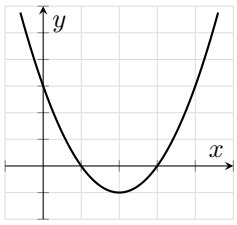
1. What the Radical in the Formula Tells Us

In $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, the value under the radical decides what kind of roots there are. If it is positive, the $+$ and the $-$ give two different numbers; if it is 0, they give the same number; if it is negative, there is no real square root.

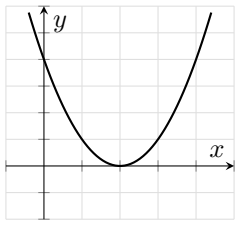
Example 1

The graphs of $y = x^2 - 4x + 3$, $y = x^2 - 4x + 4$ and $y = x^2 - 4x + 7$ are shown. Complete the table.

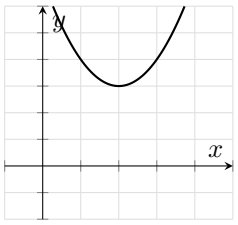
$y = x^2 - 4x + 3$



$y = x^2 - 4x + 4$



$y = x^2 - 4x + 7$



Each grid square is 1 unit.

Equation	$b^2 - 4ac$	**Roots from the formula**	**Number of real roots**	**Number of x-intercepts**
$x^2 - 4x + 3 = 0$				
$x^2 - 4x + 4 = 0$				
$x^2 - 4x + 7 = 0$				

2. The Discriminant and the Nature of the Roots

The expression $b^2 - 4ac$ is the **discriminant** of $ax^2 + bx + c = 0$. It tells the **nature of the roots** (how many real roots, and whether they are equal) without solving the equation or drawing the graph.

Discriminant	**Roots of** $ax^2 + bx + c = 0$	**Graph of** $y = ax^2 + bx + c$
$b^2 - 4ac > 0$	two distinct (unequal) real roots	two x-intercepts
$b^2 - 4ac = 0$	two equal real roots (one distinct root, a double root)	one x-intercept: the vertex is on the x-axis
$b^2 - 4ac < 0$	no real roots (also called non-real roots)	no x-intercepts

Always put the equation in standard form before reading a, b and c. "Real roots" (with no other word) means $b^2 - 4ac \geq 0$: two distinct or two equal.

Example 2

Without solving or graphing, determine the nature of the roots of each equation, or the number of x-intercepts of each graph.

(a) $3x^2 - 5x - 2 = 0$ (b) $4x^2 + 25 = 20x$ (c) $2x^2 + 3x + 4 = 0$

(d) $y = 2x^2 - 7x + 3$ (e) $y = -3x^2 + 2x - 1$ (f) $y = 0.25x^2 - 3x + 9$

(g) Determine the number of zeros of $y = 3(x + 2)^2 + 1$ in two ways: from the vertex and direction of opening, and from the discriminant.

3. Rational or Irrational Roots

When a, b and c are rational and the discriminant is a **perfect square** $(0, 1, 4, 9, 16, \dots)$, $\sqrt{b^2 - 4ac}$ is rational, so the roots are **rational**; with integer coefficients the trinomial then factors over the integers. When the discriminant is positive but not a perfect square, the roots are **irrational** and the trinomial does not factor over the integers.

Example 3

Use the discriminant to decide whether the roots are rational or irrational. If they are rational, solve by factoring.

(a) $6x^2 + 7x - 20 = 0$ (b) $2x^2 + 5x - 4 = 0$ (c) $9x^2 + 30x + 25 = 0$

4. Finding an Unknown Coefficient

Write the discriminant in terms of the unknown, then use > 0 (two distinct real roots), $= 0$ (equal roots), < 0 (no real roots) or ≥ 0 (real roots) and solve. Reverse the inequality sign when you divide by a negative number. If the unknown is in the x^2 coefficient, it cannot make that coefficient 0 (the equation would not be quadratic).

Example 4

For what values of k does $2x^2 - 6x + k = 0$ have

(a) two distinct real roots? (b) two equal real roots? (c) no real roots?

(d) Determine the equal root when $k = \frac{9}{2}$.

When the unknown is b, the discriminant contains k^2. For example, $k^2 = 36$ when $k = \pm 6$; $k^2 > 36$ when $k < -6$ or $k > 6$; and $k^2 < 36$ when $-6 < k < 6$ (the numbers between -6 and 6 are the ones whose squares are less than 36).

Example 5

(a) For what values of k does $x^2 + kx + 9 = 0$ have equal roots? Determine the root for each value. For what values of k are there two distinct real roots? No real roots?

(b) For what values of k does $kx^2 + 6x - 3 = 0$ have no real roots?

(c) For what values of k does $(k-1)x^2 + 4x + 2 = 0$ have real roots?

5. Showing That the Roots Are Always Real

To show that an equation has real roots for *every* value of a parameter, write its discriminant and show it can never be negative. A square is never negative: $(k-2)^2 \geq 0$, and $k^2 + 16 > 0$.

Example 6

(a) Show that $x^2 + kx - 4 = 0$ has two distinct real roots for every real number k.

(b) Show that the roots of $x^2 - (k+2)x + 2k = 0$ are real and rational for every integer k. When are they equal?

6. Problems: Is It Possible?

To ask whether a quantity given by a quadratic can reach a certain value, set the expression equal to that value, write the equation in standard form, and look at the discriminant: a negative discriminant means the value is never reached; zero means it is reached exactly once (at the vertex: the maximum or minimum).

Example 7

A ball is thrown upward. Its height after t seconds is $h = -5t^2 + 20t + 1$ metres. Use the discriminant to answer each question.

(a) Does the ball reach a height of 25 m?

(b) Does it reach 16 m? How many times?

(c) Determine its maximum height.

Lesson 9: Maximum and Minimum Problems: Graphical Approach

Using the graph of a quadratic model, on paper or on a graphing calculator, to find greatest and least values, starting values, landing times and the times a model reaches a given value.

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1. Reading a Quadratic Model

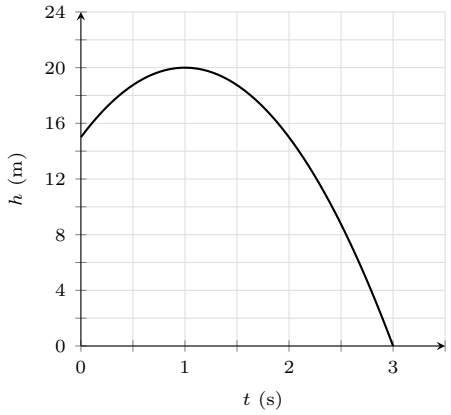
A **quadratic model** is a quadratic function that describes a real situation, such as the height of a ball, the revenue of a business or the area of a shape. Models are often written in function notation: in $h(t) = -5t^2 + 10t + 15$, the input is the time t (seconds) and the output $h(t)$ is the height (metres), so $h(2)$ means "the height after 2 seconds". Each question about the situation matches a feature of the graph.

The question asks for...	**Use this feature of the graph**
the greatest or least value (maximum height, maximum revenue, minimum cost)	the y-coordinate of the **vertex**
when or where the maximum or minimum happens	the x-coordinate of the **vertex**
the starting value (at launch, at time 0)	the **y-intercept**
when an object lands or hits the ground	the positive **zero** (x-intercept)
when the output equals a given value k	the **intersection** with the line $y = k$
the output for a given input	**substitute** (evaluate)

In a context, the **domain** and **range** are only the values that make sense. A ball's model is used from launch ($t = 0$) to landing, so the domain stops at the landing time and the range runs from 0 up to the maximum height (not from the launch height: the ball falls past it to the ground).

Example 1

A ball is thrown from the top of a building. Its height in metres after t seconds is $h(t) = -5t^2 + 10t + 15$, graphed at the right. Use the graph to answer.



(a) From what height was the ball thrown?

(b) What is the maximum height, and when is it reached?

(c) When does the ball hit the ground?

(d) At what other time is the ball at its launch height?

(e) State the domain and range of the model in this context.

2. Using a Graphing Calculator

When the vertex or the intercepts are not on grid lines, a graphing calculator finds them to as many decimal places as you need.

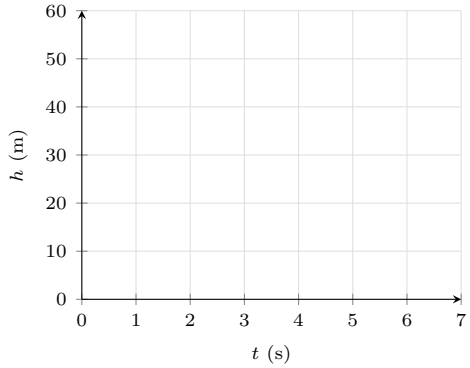
TI-84 Plus CE (the Casio has the same tools under G-Solv):
1. Press Y= and enter the model with X as the input variable (for $h(t) = -4.9t^2 + 32t + 2.5$, enter $-4.9X^2 + 32X + 2.5$).
2. Set the WINDOW from the context: start X and Y at 0, and make Xmax and Ymax a little larger than the times and values you expect. Press GRAPH.
3. Press 2nd [TRACE] (CALC) and choose 1:value (the output for an input), 2:zero, 3:minimum, 4:maximum or 5:intersect (after entering $Y_2 = k$ for a given value). For zero, minimum and maximum, move to a left bound and a right bound on either side of the point, press ENTER for each, then ENTER again for the guess.
4. Round only the final answer, as the question asks. 2nd [GRAPH] (TABLE) lists outputs for whole-number inputs.

Always sketch the graph you see, with the axes labelled (name and units) and the important points marked.

Example 2

A model rocket is launched from a platform. Its height in metres after t seconds is $h(t) = -4.9t^2 + 32t + 2.5$. Use a graphing calculator, and answer to the nearest hundredth.

(a) Sketch the part of the graph that fits the context.



(b) How high is the rocket after 2 s?

(c) What is its maximum height, and when is it reached?

(d) How high is the platform?

(e) How long is the rocket in the air?

(f) When is the rocket 30 m above the ground?

(g) State the domain and range of the function in this context.

Example 3

A power line hangs between two poles 120 m apart. Its height in metres above the ground, x metres from the left pole, is $h(x) = 0.004x^2 - 0.48x + 25$, $0 \le x \le 120$.

(a) How high is the line where it is attached to each pole?

(b) What is the lowest height of the line, and how far from the left pole is it?

(c) At what distances from the left pole is the line 15 m high? Answer to the nearest tenth of a metre.

(d) State the range of the function in this context.

3. Revenue Models

Revenue is the money taken in: revenue = (price per item) × (number of items sold). When a business raises its price, it usually sells fewer items, so the revenue can go up or down. If x is the **number of price changes**, both the price and the number sold are linear expressions in x, and their product is a quadratic.

Let x be the number of price changes. Price = (current price) $\pm$ (change) x, Number sold = (current number) $\mp$ (change in sales) x. $R(x)$ = (price)(number sold), and the vertex of its graph gives the maximum revenue.

If the price can only change in whole steps ("each \$2 increase"), x must be a whole number. When the vertex is not at a whole number, compare the revenue at the whole numbers on either side of it (the TABLE does this quickly).

Example 4

A tour company sells 600 boat-tour tickets a week at \$45 each. A survey shows that for every \$2 increase in the price, 15 fewer tickets would be sold. The price changes only in steps of \$2.

(a) Let x be the number of \$2 increases. Write expressions for the price and the number of tickets sold, and the revenue function $R(x)$.

(b) Graph $R(x)$ and find its maximum. Why is that not the final answer?

(c) Use the table to choose the best whole number of increases. State the best price, the number of tickets sold, and the maximum weekly revenue.

(d) How much more is this than the current weekly revenue?

4. Models from Data and from Geometry

Many quadratic models come from measured data: a calculator draws the "curve of best fit" (quadratic regression, a later course) and the model is then used like any other. A model only describes the situation over the inputs it was built from; it can be poor far outside them. Area problems often lead to a quadratic in one length, and then the domain is the set of lengths that make every dimension positive.

Example 5

A driver records the fuel her car uses at different steady speeds.

Speed, v (km/h)	40	60	80	100	120
Fuel use, E (L per 100 km)	6.3	5.0	4.7	5.6	7.1

The data are modelled by $E(v) = 0.0012v^2 - 0.18v + 11.5$ for $40 \le v \le 130$.

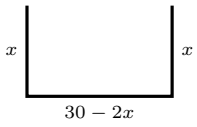
(a) At what speed does the car use the least fuel, and how much does it use?

(b) Compare the fuel use at 50 km/h and at 110 km/h.

(c) At what speeds does the car use 6 L per 100 km? Answer to the nearest km/h.

Example 6

A rain gutter is made from a strip of metal 30 cm wide by bending up x cm on each side, as shown. The area of the cross-section, in square centimetres, is $A(x) = x(30 - 2x)$.



(a) State the domain of $A(x)$ in this context.

(b) What value of x gives the greatest cross-section area? What is that area?

(c) For which values of x is the area 100 cm^2?

Lesson 10: Maximum and Minimum Problems: Algebraic Approach

Writing a quadratic function for a situation and finding its maximum or minimum exactly, by completing the square, by $x = -\frac{b}{2a}$ or from the zeros.

---

1. Finding the Vertex Algebraically

The maximum or minimum value of a quadratic function is the y-coordinate of its vertex; it happens at the x-coordinate. There are three algebraic ways to find the vertex. Use whichever suits the form of the function.

Vertex form $y = a(x-p)^2 + q$ (complete the square if needed): vertex (p, q).
General form $y = ax^2 + bx + c$: the axis is $x = -\dfrac{b}{2a}$; substitute this x to get the y-coordinate.
Factored form $y = a(x-r)(x-s)$: the axis is midway between the zeros, $x = \dfrac{r+s}{2}$; substitute.
If $a < 0$ the parabola opens down and the vertex is a **maximum**; if $a > 0$ it opens up and the vertex is a **minimum**.

Completing the square on $y = ax^2 + bx + c$ once and for all gives the **vertex formula** $\left(-\dfrac{b}{2a},\ \dfrac{4ac - b^2}{4a}\right)$. It is a shortcut when the numbers are fractions; substituting $x = -\frac{b}{2a}$ always gives the same y-value.

Example 1

Determine the vertex of the graph of each function, and state the maximum or minimum value.

(a) $y = 3(x+2)^2 - 5$

(b) $y = -2x^2 + 12x - 7$

(c) $y = (x-3)(x+7)$

(d) $y = 2x^2 + 5x - 1$

Example 2

A ball is hit upward. Its height in metres after t seconds is $h(t) = -5t^2 + 30t + 2$.

(a) Complete the square to write $h(t)$ in vertex form.

(b) How high is the ball after 1 s?

(c) What is the maximum height, and when is it reached?

(d) How high was the ball when it was hit?

(e) When does the ball hit the ground? Answer to the nearest hundredth of a second.

2. Setting Up the Function

In most problems the function is not given: you build it.

1. Let x be one quantity (draw a diagram when there is a shape). **2.** Write every other quantity in terms of x. **3.** Write the function for the quantity to be made greatest or least. **4.** Find the vertex algebraically. **5.** Check that the x-value makes sense (lengths positive, counts whole). **6.** Answer every part of the question in a sentence, with units.

Example 3

(a) Two numbers have a sum of 30. Determine the numbers whose product is as large as possible, and that product.

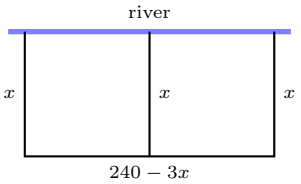
(b) One number is 4 more than three times another. Determine the numbers whose product is as small as possible, and that product.

3. Area Problems

With a fixed length of fencing, the area of a rectangle depends on its shape. Write one side as x, use the fencing total to write the other side in terms of x, and multiply. Only the sides that are fenced use fencing: a side along a wall or a river does not, and a dividing fence inside does.

Example 4

A farmer has 240 m of fencing to make a rectangular pen beside a river. No fence is needed along the river, and a fence perpendicular to the river divides the pen into two parts, as shown.
(a) Let x be the length of each fence perpendicular to the river. Show that the area is $A(x) = 240x - 3x^2$.

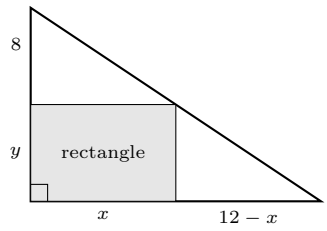


(b) Complete the square to find the maximum area.

(c) State the dimensions of the pen with the maximum area.

Example 5

A rectangle is drawn inside a right triangle with legs 12 cm and 8 cm. Two sides of the rectangle lie along the legs, and its fourth corner is on the hypotenuse, as shown.
(a) Let the width of the rectangle be x cm. Use similar triangles to show that its height is $y = 8 - \frac{2}{3}x$.



(b) Write the area $A(x)$ of the rectangle and find the dimensions that give the greatest area.

(c) Compare the greatest area with the area of the triangle.

4. Revenue Problems

Let x be the number of price changes, write the price and the number sold in terms of x, and multiply: $R(x) =$ (price)(number sold). Leaving $R(x)$ as a product of two linear factors is useful: each factor gives a zero, and the maximum is midway between the zeros.

Example 6

A theatre sells 900 concert tickets at \$40 each. For every \$5 decrease in price, it expects to sell 150 more tickets.

(a) Let x be the number of \$5 decreases. Write the revenue function.

(b) Use the zeros to find the price that gives the maximum revenue.

(c) State the number of tickets sold and the maximum revenue. Compare with the current revenue.

(d) Check part (b) with $x = -\dfrac{b}{2a}$.

5. When the Question Asks for a Given Value

Not every question in a problem asks for a maximum. "When is the ball 12 m high?" or "What are the dimensions?" means: set the function equal to the given value, write the equation in standard form, and solve it by factoring or by the quadratic formula. Two admissible answers are common (on the way up and on the way down); a negative discriminant means the value is never reached.

Example 7

(a) A stone is thrown straight up. Its height in metres after t seconds is $h(t) = 18t - 5t^2$. When is it 12 m high? Answer to the nearest tenth of a second and explain the two answers. Can the stone reach 20 m?

(b) The two shorter sides of a right-triangular garden bed have a total length of 17 m, and the longest side is 13 m. Determine the lengths of the two shorter sides.

Lesson 11: Unit 5 Review

The three forms of a quadratic function, transformations, completing the square, the quadratic formula, the discriminant and max/min problems on one page, mixed examples, and a review assignment written like the unit test.

1. Unit Summary

Form	Read directly from the equation	Find the rest
Vertex form $y = a(x-p)^2 + q$	Vertex (p, q); axis of symmetry $x = p$; opens up if $a > 0$ (minimum value q), down if $a < 0$ (maximum value q).	y-intercept: let $x = 0$. x-intercepts: solve $a(x-p)^2 + q = 0$ by square roots.
General form $y = ax^2 + bx + c$	y-intercept $(0, c)$; direction of opening from a.	Axis $x = -\dfrac{b}{2a}$, then substitute for q (or complete the square). Zeros by factoring or the formula.
Factored form $y = a(x-r)(x-s)$	Zeros r and s (x-intercepts $(r, 0)$, $(s, 0)$); direction of opening from a.	Axis $x = \dfrac{r+s}{2}$ (midway between the zeros), then substitute for q. Expand for general form.

Transformations of $y = x^2$ to $y = a(x-p)^2 + q$. $|a| > 1$: vertical stretch by a factor of $|a|$; $0 < |a| < 1$: vertical compression; $a < 0$: reflection in the x-axis. p: horizontal translation ($p > 0$ right, $p < 0$ left). q: vertical translation. Mapping: $(x, y) \to (x + p,\ ay + q)$. Step pattern from the vertex: over 1, 2, 3 ... the graph moves $1a$, $3a$, $5a$ ... vertically.
Domain $\{x \mid x \in \mathbb{R}\}$. Range $\{y \mid y \geq q, y \in \mathbb{R}\}$ if $a > 0$, $\{y \mid y \leq q, y \in \mathbb{R}\}$ if $a < 0$. Number of x-intercepts: a and q opposite signs $\to 2$; $q = 0 \to 1$; same signs $\to 0$.

Writing the equation. Vertex and a point: substitute the point into $y = a(x-p)^2 + q$ and solve for a. Zeros and a point: substitute into $y = a(x-r)(x-s)$. In a context, choose a convenient origin (at the vertex, or at one end on the ground) and say where it is.

Completing the square ($y = ax^2 + bx + c \to y = a(x-p)^2 + q$). (1) Factor a from the x^2 and x terms. (2) Inside the bracket add and subtract (half the coefficient of $x)^2$. (3) Move the subtracted number out of the bracket by **multiplying it by** a. (4) Write the perfect square and simplify. Check: the y-intercept must still be c.

Quadratic formula. $ax^2 + bx + c = 0$, $a \neq 0$: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. Give exact answers in simplest radical form, or round as asked.
Discriminant $b^2 - 4ac$: > 0: two distinct real roots (two x-intercepts); $= 0$: one real (equal) root (vertex on the x-axis); < 0: no real roots (no x-intercepts).
Choosing a method: factor if it factors; square roots for $a(x-p)^2 + q = 0$ or $ax^2 + c = 0$; the formula for anything else; graphing (zero or intersect) to check or for decimal answers.

Max/min problems. Given a model: the vertex gives the maximum or minimum (calculator: maximum/minimum); a zero gives when the value is 0; intersect with a horizontal line gives when it reaches a value. Creating a model: define the variable, write the quantity as a quadratic (area = length × width; revenue = price × number sold), find the vertex (complete the square, $x = -\frac{b}{2a}$, or midway between the zeros) and answer in context with units. State the domain and range that make sense.

2. Analyzing Quadratic Functions

Example 1

Consider the function $y = -2(x + 3)^2 + 8$.

(a) Describe how the graph of $y = x^2$ is transformed to give this graph.

(b) State each characteristic.

Vertex _____

Axis of symmetry _____

Direction of opening _____

Maximum or minimum value _____

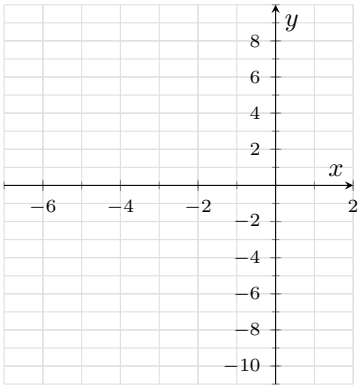
Domain _____

Range _____

y-intercept _____

x-intercepts _____

(c) Sketch the graph using the step pattern.



Example 2

The function $y = 2x^2 - 12x + 10$ is in general form.

(a) Write it in vertex form by completing the square.

(b) Write it in factored form.

(c) Use the most convenient form to state the vertex, the axis of symmetry, the y-intercept, the zeros and the range. Check the axis with $x = -\dfrac{b}{2a}$.

3. Writing the Equation of a Quadratic Function

Example 3

Write the equation of each quadratic function.

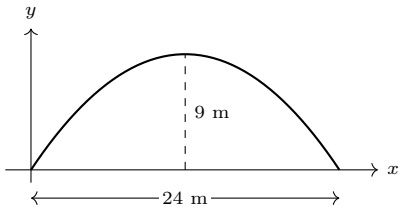
(a) Vertex $(2, -5)$, passing through $(4, 3)$. Use vertex form.

(b) Zeros -1 and 5, passing through $(3, 16)$. Give factored and general form, and the vertex.

Example 4

A parabolic arch over a park entrance is 24 m wide at the ground and 9 m high at its centre.

(a) Place the origin at the left end of the arch on the ground. Write the equation of the arch in factored form.



(b) How high is the arch 3 m in from one end, to the nearest tenth of a metre?

(c) Write the equation of the same arch with the origin on the ground directly below the top of the arch.

4. Solving Quadratic Equations

Example 5

Solve. Choose an efficient method and say which one you used.

(a) $3(x-2)^2 = 27$

(b) $2x^2 + 3x = 2$

(c) $x^2 - 6x + 4 = 0$ (exact)

(d) $3x^2 + 2x - 4 = 0$ (to the nearest hundredth)

Example 6

(a) Use the discriminant to determine the nature of the roots of (i) $4x^2 - 12x + 9 = 0$ and (ii) $2x^2 + x + 5 = 0$.

(b) For what values of k does $x^2 + kx + 9 = 0$ have equal roots? two distinct real roots?

(c) For what values of k does $kx^2 - 6x + 3 = 0$ have no real roots? Describe the graph of $y = kx^2 - 6x + 3$ for these values.

5. Maximum and Minimum Problems

Example 7

A ball is thrown upward from a balcony. Its height, h metres, after t seconds is $h = -4.9t^2 + 19.6t + 1.5$. Use a graphing calculator (graph the function, then use maximum, zero and intersect).

(a) What is the maximum height, and when is it reached?

(b) When does the ball hit the ground, to the nearest hundredth of a second?

(c) For how long is the ball higher than 15 m, to the nearest hundredth of a second?

(d) State the domain and range of the function in this context.

Example 8

A theatre sells 400 tickets a night at \$12 each. A survey shows that for every \$0.50 increase in price, 10 fewer tickets are sold. What price gives the maximum revenue, and what is that revenue?